



The Sharing Economy Globalization Phenomenon: A Research Agenda

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ABSTRACT

Sharing economy firms have been able to achieve global levels of success at an unprecedented pace. In this study, we focus specifically on internet-based firms that allow rent appropriation from temporary utilization of underutilized assets. By looking at these firms' main characteristics and the current dynamics revolving around their internationalization process, we develop a framework to guide future research drawing from a business ecosystems perspective. The business ecosystem approach is a promising theoretical lens to assess this phenomenon due to its more holistic view of multisided network effects and multiple stakeholders' participation that can vary across nations. We note that the sharing economy phenomenon has spurred worldwide adoption of platform businesses which, in turn, creates a great opportunity for future research to extend current theories by exploring why, when, and how these firms expand into new countries. We argue that further research on sharing economy firms and its interactions with different national ecosystem configurations can provide important insights to theory as well as relevant information to managers and policymakers.

1. Introduction

The sharing economy phenomenon proved to be more than just a frail and temporary trend and has been capable of overturning competition across the globe. Within the wide variety of organizations that have flourished based on the concept of “sharing” underutilized assets, a group of internet-based platform businesses have been thriving in many different industry sectors and internationally (Belk, 2014; Möhlmann, 2015). A great deal of attention has been drawn to not only their capacity to raise large amounts of funding, but also to their impressive international expansion. Relying on a virtual platform to conveniently connect users and providers, these firms have achieved instant global adoption, which has resulted in clashes with local competition and regulators. Considering the particular way in which internet-based firms create and capture value (Alcácer et al., 2016; Amit and Zott, 2001; Brouthers et al., 2016; Gawer and Cusumano, 2014), questions on how sharing economy firms formulate and implement internationalization strategies—specifically with regard to local adaptation, internalization, and competitive dynamics—still need to be addressed by the literature. In this study, we depict the nature and internationalization process of these firms, contrasting it with current international business (IB) theoretical propositions and pointing out opportunities for future research drawing from the business ecosystems approach.

The popularized “sharing economy” term has been used frequently to describe different organizations that connect users/renters

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and owner/providers through consumer-to-consumer (C2C) (e.g., Uber, Airbnb) or business-to-consumer (B2C) platforms, allowing rentals in more flexible, social interactive terms (e.g., Zipcar, WeWork). Other terms such as “gig economy,” “collaborative consumption” (Botsman and Rogers, 2010), and “access-based economy” (Bardhi and Eckhardt, 2012) have also been used to characterize these for- and nonprofit organizations. Noticing that most firms and providers in the sharing economy charge for their services, Eckhardt and Bardhi (2015) argue that the term “sharing” should be changed to “access.” In this context, given the lack of consensus regarding the sharing economy terminology, we explain in the following section our choice of focusing only on internet-based, social interactive platforms targeting rent appropriation from transactions involving temporary access to underutilized assets.

Considering the particular business model, internationalization process, and competitive dynamics of sharing economy firms, we acknowledge that future research on their global strategy implementation and interactions with countries' product/market regulatory environments could benefit both theoretical and public policy developments. The “asset-lite” nature of internet platforms and their instant global adoption pose challenges to current “asset-based” internationalization theoretical lenses drawing from transaction costs theory, such as the ownership, location, internalization (OLI) paradigm, the sequential notion of internationalization, and other “resource-based” perspectives (Bartlett and Ghoshal, 2002; Dunning, 2003; Johanson and Vahlne, 1977; Knight and Cavusgil, 2004). In this study, we address the sharing economy phenomenon from an international business standpoint, focusing on how their internationalization significantly challenges current theoretical assumptions, how existing theories could be extended from it, and proposing new avenues for future research. We also provide evidence that sharing economy firms engage in an ecosystem-based form of international expansion that is novel and needs further research to be explained. Ecosystems are seen as a community of stakeholders including firms, governments, and other players that are mutually dependent for their effectiveness and survival (Rong and Shi 2014; Autio and Thomas, 2014; Iansiti and Levien, 2004; Moore, 1993, 2006). In this context, the process of entering new international ecosystems comprises the co-evolving interaction of the focal firm with a multitude of relevant local players (Rong et al., 2015).

We explore particularly how sharing economy firms have been able to create and appropriate value both at home and abroad and deal with international competitors, complementors, customers, and regulators. We highlight that since internet platforms have reduced transaction costs toward tangible assets management due to traded assets not being internalized, they are able to internationalize at a greater speed by relying on local providers for business adaptation and focusing on efficient operational integration. We explore also how organizations and regulations within the sharing economy context influence (and are influenced by) internationalization. Moreover, we provide a framework with the purpose of serving as a starting point for future research to expand current IB theories using the sharing economy phenomenon.

In contrast to traditional B2C models, sharing economy firms can take the form of a multisided platform business (Eisenmann et al., 2011; Iansiti and Levien, 2004; Normann and Ramirez, 1993) that relies on information technology as a springboard to allow different providers to rent their assets and users to find substitute alternatives to ownership. Therefore, sharing economy platforms go beyond the sequential notion of value chains (Porter, 1985). It involves collaborative arrangements among suppliers, complementors, and users as the roles of suppliers and customers become blurred and an economic ecosystem is created based on users' networks formed in marketplaces around the globe. Sharing economy firms have expanded internationally through an aggressive market-seeking approach not bounded by cultural or economic discrepancies. These firms have relied on the social support and involvement achieved by the sharing economy phenomenon, embracing a myriad of global users and expanding internationally at unprecedented speed. However, despite the global adoption of these platforms, sharing economy firms have faced fierce competitive responses from established business and emerging local sharing economy firms. These tensions and competitive dynamics provide an exceptional setting for new theoretical developments from an international business standpoint.

We first provide a view on the nature of sharing economy firms and how they differ from other business models. Second, we show how current international business theories help us understand (and fall short in explaining) this phenomenon, contrasting these theories with sharing economy firms' internationalization process and pointing out how current IB theories could be expanded by the investigation of these firms. Finally, we propose a set of future research avenues that should be explored in order to further understand the relevance of the sharing economy phenomenon to IB literature.

2. A global perspective on the sharing economy phenomenon

Customers are increasingly demanding complex, sustainable, and integrated solutions rather than standardized and homogeneous products and services. In addition, a confluence of important events spurred the rapid development and adoption of the so-called “sharing economy.” Factors such as the ubiquity of the internet and mobile devices, the abundance of goods in idle capacity, growing consumer awareness regarding environmental sustainability, and the economic recession leading to higher unemployment rates have attracted consumers to transactions that allow them to more conveniently access and profit from underutilized assets. Global sharing economy firms, such as Uber, Airbnb, Lyft, and WeWork, have achieved impressive results both locally and internationally. These are for-profit organizations that have been undertaking aggressive competitive moves in order to gain market share both in the home country and abroad. Sharing economy firms can be seen as examples of platform capitalism, which consists of intermediary firms relying on data analytics and market coordination to appropriate rent from transactions (Langley and Leyshon, *in press*; Sundararajan, 2016). Customers also do not engage with sharing economy firms just to experience a new social interaction, but mainly to avoid ownership liabilities and take advantage of lower costs (Eckhardt and Bardhi, 2015).

Despite ongoing debate around the sharing economy term, a definition of it was added to the Oxford English Dictionary in 2015 as follows: “an economic system in which assets or services are shared between private individuals, either free or for a fee, typically by means of the Internet.” This definition highlights two important features we use in this study to characterize sharing economy firms:

Table 1

The three main features of sharing economy firms.

Characteristic of sharing economy	Uber	Airbnb	Zipcar	Netflix	eBay	Kickstarter
The business focuses on unlocking the value of unused or underutilized assets.	Yes	Yes	Yes	No	Yes	No
Consumers pay for temporary access instead of ownership using an internet-based platform.	Yes	Yes	Yes	Yes	No	No
It relies on C2C interactions and network effects for growth.	Yes	Yes	No	No	Yes	Yes

(1) assets or services are shared between individuals for a fee; and (2) transactions are mediated through the internet. In this study, we approach our view on the sharing economy close to Oxford Dictionary's definition since our objective is to explore the unique features of C2C sharing-based internet platforms that have not been explored in the literature. Thus, in our conceptualization, we highlight that in the sharing economy, assets transacted between users/suppliers are for temporary use. In sum, we set the boundaries of the sharing economy business model around the following three pillars: (1) the business focuses in unlocking the value of unused or underutilized assets; (2) consumers pay for temporary access instead of ownership using an internet-based platform; and (3) it relies on network effects and social interactions between users/suppliers for growth. As we will explain, firms having these unique characteristics are still unexplored in their internationalization process and constitute an important phenomenon for future IB research due to their “asset-less” business model based on open online platforms.

Incipient research on the sharing economy, however, has usually taken a broader approach regarding sharing-based business models, also contesting the use of the word “sharing” and introducing overarching and complementary terms such as “access economy,” “collaborative consumption,” and “on-demand services” (Bardhi and Eckhardt, 2012; Belk, 2014; Botsman and Rogers, 2010; Rauch and Schleicher, 2015). Access economy, for example, encompasses B2C businesses such as Zipcar and WeWork, which have developed a business model where the firm's assets are temporarily rented to customers who actively participate in the business maintenance and use an internet platform that gives them more flexibility over the rental terms. In our assessment, we do not include crowdfunding platforms—although they are sometimes referred to as part of the sharing economy—since they do not grant temporary access to users and do not allow rent appropriation from underutilized assets. Table 1 summarizes how different prominent internet-based platforms and rental service firms fit in our categorization of the sharing economy.

Now that we understand that sharing economy firms in the context of this study are also internet-based C2C platforms, it is important to highlight how dependent their success is on social interactions and network effects. The concept of business platforms has been investigated broadly under an economic and operations management perspective (Armstrong, 2006; Cusumano and Gawer, 2002; Eisenmann et al., 2011; Evans, 2003; Gawer, 2009; Meyer and Lehnerd, 1997; Parker and Van Alstyne, 2005). It relies on the convergence of products, services, or technologies that are independently developed by different firms around a single device or marketplace created to host and process transactions between users and providers (e.g., Windows and Linux operating systems, iTunes, Android, etc.) (Armstrong, 2006; Cennamo and Santalo, 2013; Evans, Hagi, and Schmalensee, 2008). Sharing economy firms function as an interface connecting different groups of users that interact through a virtual marketplace accordingly to certain “rules” and guidelines (Gawer and Cusumano, 2014). One key feature of internet-based platforms is the significant reduction in transaction costs related to matching a customer's need with its potential providers (Eisenmann et al., 2008; Gawer, 2009). In the case of sharing economy firms, these are specialized online, data-driven platforms targeted at reducing users' search and information costs by increasing their capacity to monitor transactions (e.g., GPS tracking) and evaluate other inherent risks (e.g., customers'/providers' reviews and ratings) while also allowing for short-term, low commitment contracts. This is possible only due to technological improvements that enable consumers and providers to meet and communicate in virtual marketplaces.

Thus, in contrast to traditional firms that focus on developing and commercializing products accordingly to a specific target market's preferences, sharing economy firms concentrate their efforts on providing a virtual marketplace that efficiently and conveniently connects users to providers, allowing them to match accordingly to time, place, and contract customizations. These firms leverage on the aforementioned benefits to grow as more users from both supply and demand sides adopt the platform (Altman et al., 2014; Eisenmann et al., 2006; Gawer, 2009). Sharing economy platforms capture value by charging a fee for each transaction made through their platform and depend on widespread popular adoption and participation to succeed (Eisenmann et al., 2008; Hagi and Altman, 2017). As more asset owners and service providers make their resources available on the platform, consumers are attracted by price reductions and increased customization options. Thus, sharing economy platforms are highly influenced by network effects, which occur when the value of a product or service increases as more users utilize the platform and expand the network (Abrahamson and Rosenkopf, 1997; Evans et al., 2008; Gawer and Cusumano, 2014; Katz and Shapiro, 1985).

These platforms are “open” on the supply side, which means there are reduced barriers for suppliers to join (e.g., little restrictions regarding technical standards) and greater alternatives for users to customize the service according to their needs (Eisenmann et al., 2008; West, 2003). By focusing on the ownership of the virtual platform, sharing economy firms have their core competence based on the digital intermediation of the value chain. These firms focus their resource allocations into marketing efforts and operational efficiency based on superior data analysis, as assets being transacted are not internalized. This business model makes sharing economy firms' organizational structures simple, mainly divided into platform technology, operations, marketing, and customer service. (Law services, an important complementary activity, are sometimes internalized.) For example, Uber, which operates in more than 60 countries and had total revenues of more than \$1.5 billion in 2015, coordinates all operations such as software development, marketing, and legal issues from its San Francisco headquarters. In the case of Airbnb, it has its value creation and capture based on its capacity to access, store, and provide information regarding lodging solutions to consumers, while also serving as a global

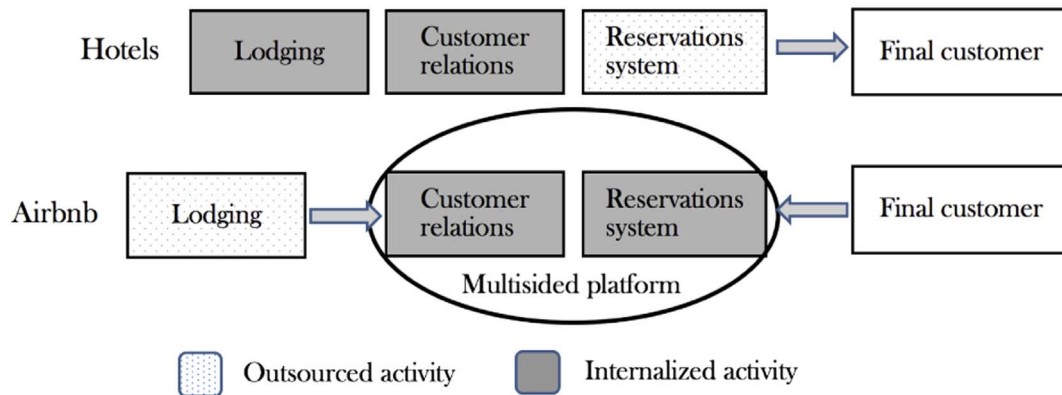


Fig. 1. Differences between sharing economy firms and traditional businesses' organizational boundaries.

marketplace for providers to display their offers. Thus, the main difference in terms of organizational structure between Airbnb and hotel chains is that Airbnb does not own lodging locations but owns the reservations system, while hotels own properties but rely heavily on search engines for reservations. Airbnb focuses on having internalized only the value chain part (the reservations system and customer relationship management, as shown in Fig. 1), which is a key characteristic of sharing economy firms.

Since sharing economy firms do not possess rare, difficult to imitate assets to be used as a source of competitive advantage both at home and abroad, attaining first-mover advantages is of paramount importance. In the case of Uber, executives have pointed out that investing their marketing efforts in geographically dense locations worldwide is crucial for the creation of a “virtuous cycle” in which network effects are exponentially expanded (Radhakrishnan, 2015). Since sharing economy firms utilize local user's resources to supply local demand, their costs to expand into other countries are reduced when compared to “asset-heavy” business models. Sharing economy firms' most costly investment in international expansions is the platform coding configuration for local adaptation. Radhakrishnan (2015), former head of product at Uber, acknowledged publicly that “the first time that we translated uber.com, we realized that every change in content required multiple code changes and redeploying the entire website each time.” Product adaptations at the platform level are also required in some cases, for example, taking cash as payment and developing a lighter app for users possessing slow, low-end devices.

Sharing economy firms have expanded internationally at great speed due to their asset-lite business model that leverages from global adoption of the internet and mobile devices. In this context, trust among users is of paramount importance to sharing economy firms. Free communication between users and providers is encouraged to happen through the platform only, and open rating systems allow users to better evaluate their options. Heavy investments are made to ensure customers that utilizing a sharing-based service through their platform is as safe as contracting from any other traditional firm. For example, boat sharing firm Boatsetter, currently operating in more than 20 countries, integrates boat captains in the platform besides customers and providers. This addition not only provides both users and owners a trustworthy setting for transactions, but also increases network effects due to multisided interactions and propagation of the platform. By introducing captains to the platform, boat owners could suggest captains they trust and users could more conveniently search for a captain. On the other end, captains would benefit from more work opportunities. Boatsetter also provides insurance of up to \$1million to all rides, thus reinforcing trustworthiness so as to attract more boat owners and users to the platform.

The main inhibitors of sharing economy firms' international expansion are lack of technology infrastructure, lack of complementary asset providers, and local regulations, which play a key role in this ecosystem. Sharing economy firms need to carefully manage their strategic alliances with complementary asset providers in order to facilitate entry on both supply and demand sides and increase their ecosystem dominance. Uber's partnership with complementary asset providers, such as Hertz and Toyota, in order to stimulate drivers to rent or lease new cars exemplifies this strategic behavior. Moreover, although it is easier for these firms to initiate operations abroad compared to their rivals from traditional business models, they have struggled with local regulatory systems and local emerging sharing economy platforms. Questions regarding liabilities, consumer protection, and labor conditions have spurred a lot of debate and tension between incumbent players, regulatory institutions, and the sharing economy ecosystem. So far, it appears that although sharing economy firms have lost some legal battles to regularize their business in some countries, they have been successful in most nations, and it should be only a matter of time until currently blocked locations are open—probably under stricter rules—for sharing economy firms to operate due to strong popular pressure. The internationalization pattern of sharing economy firms and the battles with local players and regulators reveal that their expansion strategy prioritizes market share gain and first-mover advantages despite local issues related to competitive retaliation and regulatory enforcement (Cennamo and Santalo, 2013). That is because the success of such platform-based business model depends on widespread users' adoption, which is leveraged by the “buzz” it creates. For example, Uber has been noticed to pursue a “shock and awe strategy,” which consists of drawing a great deal of attention and generating consumer enthusiasm which, in turn, helps the firm in its lobbying efforts to regularize its operations (Abboud and Wagstaff, 2015).

Recent reports (see Fig. 2) have shown that people in Asia-Pacific and Latin American regions have a greater willingness to engage in the sharing economy than those in North America and Europe (Nielsen, 2014). This increased willingness to participate in the

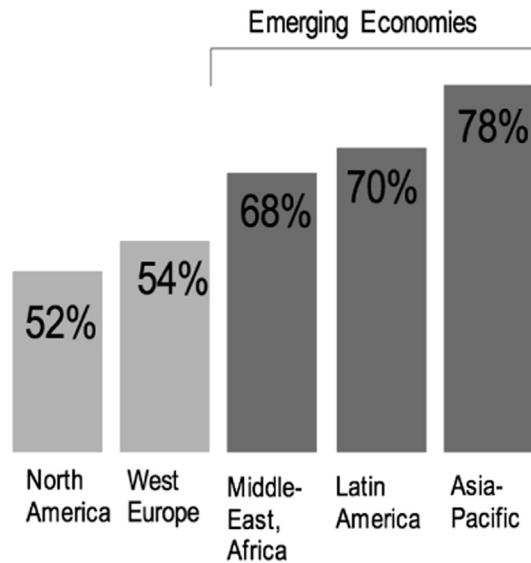


Fig. 2. Willingness to participate in the sharing economy.
Source: Nielsen (2014)

sharing economy in emerging markets is fueled by consumers' perception of greater value added when dealing with more transparent, fairer transactions. For example, Ozimek (2014) points out that since emerging markets usually have institutional and informational voids and high levels of bureaucracy and doing business costs, sharing economy services become very appealing to individual providers and become legitimated in the market. Despite resistance and retaliation from incumbent firms and the local government, popular pressure has fiercely backed up sharing economy firms in emerging markets. Firms like Go-jek, an Indonesian "Uber for motorcycles," Airbnb entrepreneurial hosts from "slums" in Rio de Janeiro, and PrepClass, a Nigerian educational platform, together with crowdfunding platforms have created thousands of jobs and fomented many new ventures in these emerging markets. Although sharing economy firms stimulate freer market transactions, this does not mean that it should work only in unregulated markets. Local governments around the globe have been imposing strict rules to shared-based transactions and operations (Osarenkhoe, 2014). Since sharing economy firms may take advantage of price elasticity when demand is high and supply is low, regulators and insurance companies face a complex issue of regulating customers' and providers' rights and liabilities. Such regulations are an important step toward legitimating the sharing economy business and protecting consumers' rights and employees/providers fair working conditions.

Another impact sharing economy firms brought to global businesses is the increase of local and international competition. For example, information technology (IT) and software development experts from different nations are available to work as independent service providers online, and they can be contacted, evaluated, and contracted via sharing economy platforms such as [freelancer.com](https://www.freelancer.com). Therefore, by relying on a myriad of local individual providers, sharing economy firms have quickly become giant global contenders. The success of sharing economy firms has also been causing disruption and increasing competition in many different industries. The recent movement toward transforming traditional business into platform-based, data-driven marketplaces has been the most prominent business model among start-ups. These entrepreneurs have noticed that the online platform concept can be applied to other businesses, such as real estate, home delivery, and services in general.

Local and internationalized firms have started to adjust their strategies regarding pricing, customization, and differentiation. Hotel chains have been investing in their own online search and price match engines and also developing new concepts such as the CitizenM and Pod Hotel, which are targeted to potential sharing-economy customers (Edelman and Geradin, 2016). The global automotive industry is also making important strategic moves in order to adapt to the sharing economy. Toyota has acquired a minority stake in Uber and has created leasing offers to Uber drivers. General Motors has invested in Lyft, Uber's main competitor, allowing drivers to lease their cars at better rates. Volkswagen, Daimler, and BMW have all invested in car-sharing companies as well, indicating that these firms are exploring the opportunity to partner with sharing economy firms instead of competing directly against them. By looking at these competitive and cooperative dynamics, we can observe that incumbent firms facing sharing-based competition are challenged to pursue at least one of these options: (1) make important organizational and operational modifications in order to reduce costs, such as in the case of local taxi fleets and hotel chains; (2) redesign or develop a new business model that is more flexible and customizable or acquire a sharing-based business, such as the case of Avis' acquisition of Zipcar; and/or (3) join forces with sharing economy firms and look for synergies, as exhibited by the example of automakers. In sum, as the sharing economy gains momentum in different industry sectors, it will also spur strategic changes and competitive moves by incumbent firms.

Competition between local and foreign sharing economy firms is also an important factor influencing their internationalization process. For example, Uber's sell off of its Chinese business to Didi reveals that late entry and lack of adaptation in nurturing the ecosystem (Rong et al., 2015) can play a significant role. Didi aggressively gained the majority of the market share by getting support

from large reputable national firms (such as Alibaba and Tencent), merging with another ride-hailing company (Kuaidi), and incorporating payments and chat services from the most popular chat app in China (WeChat); Uber tried to replicate its app without any significant changes. Uber was losing so much money in China that it ended up draining its global profitability (Newcomer and Wang, 2016). Failing to acknowledge local users' preferences in the national ecosystem, underestimating local competition, and avoiding local partnerships has put Uber alongside eBay and Yahoo as examples of unsuccessful internationalization in the Chinese market.

Drawing from the aforementioned description of sharing economy firms and its business model characteristics and impacts on industry competitive dynamics, we now turn to a theoretical discussion of how current IB theories can be expanded through the investigation of these firms' internationalization. Next, we explain how the sharing economy business model and its internationalization process contrast with current theoretical knowledge on international business.

3. How do current IB theories explain sharing economy firms' internationalization?

Since its early stages, the IB literature has relied on the premises of transaction cost economics (Coase, 1937; Williamson, 1975). Deriving from the notion of “the nature of the firm,” early IB scholars extended to an international landscape the view that firms exist as means of reducing transaction costs via internalization and coordination of imperfect markets (Caves, 1971; Dunning, 2003; Hymer, 1976; Vernon, 1966). From this perspective, multinational enterprises (MNEs) started to be studied as organizations with internalized value-adding activities able to exploit firm-specific and country-specific advantages overseas. Firms then could explore, exploit, and appropriate value from their specific tangible or intangible assets only via internalization, which is determined by cost-reduction decisions of internalizing activities up to the point where the costs of further internalization outweigh the benefits (Buckley and Casson, 1976; Kirca et al., 2011).

Drawing from the internalization concept, the OLI or eclectic paradigm developed by Dunning (1980) proposed that coupled with internalization (I), ownership (O), and location (L), advantages are necessary for firms to engage in cross-border activities. A great number of studies have empirically tested the application of these theories and concluded that internalization is an indispensable element of internationalization (Cuervo-Cazurra, 2012; Kirca et al., 2011), while the OLI paradigm is the most predominant and comprehensive theoretical approach explaining foreign direct investment. However, when we contrast the nature and internationalization process of sharing economy firms with internalization-based theoretical lenses, we note some significant differences. Since the internationalization behavior of internet-based firms has already been discussed and contrasted with past IB theories (Brouthers et al., 2016; Kotha et al., 2001; Singh and Kundu, 2002), we avoid the overlap and focus on key theories that might be expanded and also serve as a basis for future sharing economy internationalization research.

Starting with the eclectic paradigm, we highlight that O advantages are bounded by the virtual platform value chain since most of the assets are owned by users which, in turn, reduces issues of reallocation/coordination of production and product/service local adaptation. In this case, as platform-based firms strive to leverage network effects by increasing both users' and suppliers' adoption, O advantages are built around intangible assets more ready to be redeployed internationally, such as marketing capabilities, superior technology, and intellectual property (IP). These O advantages also provide sharing economy firms with greater capacity to speed internationalization, since guaranteeing first-mover advantages is crucial to these firms. Yet, the reputation and brand image created by sharing economy firms is of fundamental importance for their widespread adoption. Therefore, these firms invest heavily in operational efficiency and technologies to leverage reputation and secure their position in the local market. However, we highlight that sharing economy firms' O advantages are not rare or hard to imitate and, therefore, are not their only source of competitive advantage.

In the case of L advantages, we observe that even though sharing economy firms have been impacted by constraints regarding technological infrastructure and market regulations, it has not prevented them from entering markets where overall L advantages are low but the market is attractive. However, sharing economy firms' costs for legal fees and lobbyists have been substantial and a major issue when the firms enter new markets where regulations are not easily updated or where established local firms have increased influence on the political system. Subsidiaries are provided only to promote marketing activities and reinforce operational efficiency, while in some cases also to avert taxation. That indicates that sharing economy firms do not rely on location-bound endowments to go abroad (usually related to an asset-seeking approach), and instead have a pure market-seeking interest.

With regard to I advantages, sharing economy firms do not necessarily need to internalize foreign activities in order to reduce transaction costs. On the contrary, these firms target “creating” and nurturing a marketplace where upstream and downstream players own the assets being transacted. Thus, sharing economy firms do not rely on tangible internalized assets across borders to create and capture value, which contrasts what strategy and international business theories have considered as important sources of competitive advantage (Kirca et al., 2011; Porter, 1990; Verbeke, 2003). Considering that sharing economy firms' firm-specific advantages are network- and technology-based, internalization advantages are based on knowledge and technological capabilities are utilized to propagate network effects. In this case, these firms do not have incentives to vertically integrate operations, as their core competence lies in the management of the virtual platform, which requires the internalization of technologies and services owned by complementary asset providers. In sum, the OLI paradigm is a good starting point to study sharing economy firms and, considering their particular nature that allows reduced internalization needs, intangible assets-based ownership advantages, and less dependency on location advantages, we notice that further investigation on how sharing economy firms manage their internationalization could significantly expand the OLI perspective.

Coupled with the eclectic paradigm, we also assess how other theoretical perspectives can be useful to the investigation of sharing economy firms and which ones are not applicable. We start by noticing that sharing economy firms do not have to move production

abroad to reduce costs or source cheaper inputs, and there is no need to wait for product standardization to move into less developed economies, as opposed to what is predicted in the product life cycle model (Vernon, 1966). Thus, some of the main reasons for international expansion—such as exploitation of economies of scale and scope, lower input costs, and market imperfections (Buckley and Casson, 1976; Porter, 1990; Vernon, 1966)—fall short in explaining the internationalization of sharing economy firms. The C2C platform business model allows them to bridge cultural gaps with greater ease due to internet globalization, whereas important information is always readily available and rating systems increase trustworthiness of transactions. An interesting point brought up by sharing economy executives is that international location choices are made primarily based on first-mover advantage and market size. Uber executives have argued that top priority international locations are selected based on population size and internet penetration in order to develop a “virtuous cycle” of increasing demand, drivers, and reinvestments through expansion into new markets. In this case, since sharing economy firms are usually start-ups that need to attract large amounts of funding to sustain their operations, internationalization becomes a necessity due to pressures for growth based on the network effects involved and first-mover advantages to be seized. This, in turn, creates a motivation for internationalization that is not only based on search for new markets, but actually a necessary condition for survival.

Yet, sharing economy firms have demonstrated fewer concerns toward some hurdles of international expansion such as “psychic distance” effects as well as institutional and cultural differences. These firms have significantly reduced entry and adaptation costs since they have greater advantages of local adaptation for using local providers' assets and resources. This pattern contrasts what incremental internationalization studies have postulated (Anderson and Gatignon, 1986; Johanson and Vahlne, 1977), as MNEs should make their country selections based on risk aversion and learning effects and go abroad incrementally by choosing to learn from closer operations first. This also means that the view of multinational firms as orchestrators of various international units that need local responsiveness and different levels of integration and are established according to product, market, and cost configurations (Prahalad and Doz, 1987) is not accurate for sharing economy businesses.

However, even though sharing economy firms may be able to launch operations across borders with greater ease, issues associated with the liability of foreignness and outsidership (Brouthers et al., 2016; Calhoun, 2002; Zaheer and Mosakowski, 1997) will still have an important role in their success. As seen by the example of Uber's failed attempt at conquering the Chinese market, sharing economy firms may have to carefully adjust not only their platforms, but also their relationship with key stakeholders in the ecosystem. Local emerging sharing economy firms can rapidly copy successful platform concepts from abroad and implement them in the home country with enhanced features customized to the local market. Yet, since these platforms depend on network effects across different players in the ecosystem, the liability of foreignness becomes even more complex. In contrast to traditional businesses that have the option of internalizing upstream and/or downstream activities in order to improve efficiency and reduce costs, sharing economy firms' platforms may have to work around other alternatives such as involving complementary asset providers and partnering with established reputable firms, like in the case of Uber's partnerships with Pepsi, Amex, and Hilton.

Another relevant IB perspective to be contrasted with sharing economy firms is the concept of “born globals.” These small firms establish international commercialization and operations within their first years of founding (Knight and Cavusgil, 2004). Sharing economy firms have many aspects in common with born globals, such as early internationalization, innovativeness, and entrepreneurial origins. Our perspective is that since the born globals research has encompassed only small, funding-constrained firms that rely on unique products and capabilities to be exported (Coviello, 2015; Knight and Cavusgil, 2004; Oviatt et al., 1995), the investigation of sharing economy firms as born globals could bring significant contributions to this theory: sharing economy firms' business models, internationalization processes, and motifs are quite distinct compared to exporting firms. Although research on born globals is still incipient (Zander et al., 2015), it could provide a good basis for the investigation of sharing economy firms with respect to the entrepreneurial orientation and environmental conditions necessary to foster such businesses.

From an institutional perspective, past studies have highlighted that MNEs face the tension between having global operations homogeneously integrated and adapting subsidiaries to local institutional practices (Kostova and Roth, 2002). Since sharing economy firms have their organizational structure heavily centered in the virtual platform, their local adaptation to new institutional environments is directly linked to the headquarter, which increases the risks of delayed response and liabilities of outsidership (Brouthers et al., 2016; Johanson and Vahlne, 2009). By seeing global sharing economy firms as drivers of global knowledge networks (Cantwell, 2013; Cantwell and Mudambi, 2005), we also highlight that sharing economy firms have demonstrated a stronger capacity to spur local institutional changes due to the large-scale adoption of internet-based business. This indicates that sharing economy firms might be more than just passive actors embedded in local institutional environments. Institutional theory could also help explain the combination of home country normative, regulatory, and cognitive aspects necessary to foster the generation and adoption of “sharing-based” business models (Geels, 2004; Kostova, 1997; Scott, 2014).

Finally, we note that sharing economy firms' internationalization is an enhanced representation of what previous studies have theorized on the effects of information and communication technologies (ICTs) on firm internationalization (Alcácer et al., 2016; Brouthers et al., 2016; de la Torre and Moxon, 2001). Departing from the assumptions of the “information age” and internationally networked capitalism (Brouthers et al., 2016; Dosi et al., 2013), we highlight that ICTs have laid the foundation for sharing economy firms' business models, allowing these firms to navigate through different international markets at greater speed.

In sum, despite the still incipient internationalization process of sharing economy firms, it is clear that their business models and behaviors overseas have many distinct aspects related to current IB theories. Do we need a new theory for the phenomenon of global sharing economy firms? Our viewpoint is that current IB theories are a useful starting point to explain the internationalization process of sharing economy firms, however they need to be extended by further theoretical developments and empirical investigations. Drawing from Cuervo-Cazurra's (2012) assessment of IB theories, we organized Table 2 to point out how sharing economy firms' internationalization process contrasts with current IB main theories. The table also points out some future research questions, which

Table 2

How sharing economy firm internationalization contrasts with IB theories.

Theory	Tenets	How sharing economy firms differ	Questions for future research
Product life cycle	As product and production become standardized, MNEs expand abroad to reduce costs of production and access new markets	Sharing economy firms do not need to move production; sharing economy firms have rapid global adoption, particularly in emerging markets	How do sharing economy firms foster adoption of their platform in different national settings? How they reduce costs in poorer nations?
Incremental internationalization	Foreign expansion should be implemented incrementally, drawing from past experience in order to reduce risks and psychic distance barriers	Sharing economy firms' location choices are based on market potential, with little concerns about psychic and/or institutional differences	How do customers' and suppliers' network effects contribute to sharing economy? How do sharing economy firms spur such effects in different locations?
OLI framework	MNEs utilize home-based owned assets (O) coupled with location-bound (L) characteristics and internalized activities (I) to obtain competitive advantages abroad	O advantages are based on knowledge/technology that can be redeployed virtually everywhere; L advantages are ecosystem based; and I advantages are technology related	What ownership advantages do sharing economy firms need to differentiate them from other platforms? How do sharing economy firms interact with the local ecosystem to attain competitive advantages?
Internalization theory	MNEs have to internalize cross-border activities as international arm's-length transaction costs increase and hamper value capture	Internalization options are limited to platform operations; boundaries become blurry as upstream and downstream markets are fostered, not internalized	What, how, and when should sharing economy firms internalize operations when expanding abroad?
Integration/responsiveness and legitimization models	MNEs have to adjust their boundaries and levels of control overseas in order to respond to local conditions and achieve legitimization	Sharing economy firms rely on virtual platforms that are globally integrated; local adaptation may not lead to organizational change	How do MNEs migrate from global product-based to global platform-based business? How does customer behavior influence the organizational design and business models of sharing economy firms in different national ecosystems?
Resource-based view and knowledge-based view	Firm-specific, rare, and difficult to imitate resources and superior knowledge management are responsible for MNE success abroad	Tangible resources are substituted by intangible ones; competitive advantages stem from widespread adoption, ecosystem dominance, and trust instead of rarity or difficulty to imitate/substitute assets	What challenges do MNEs face and what strategies are developed to differentiate themselves and compete against other MNEs and other sharing-based business?
Institutions-based view	MNEs adjust their operations according to local institutions and have their internationalization process shaped by home country imprinting effects	Local regulations are a constraint, but are being shaped by popular pressures; C2C commerce stimulates free market reforms	What is the role of the local regulatory environment in shaping (being shaped by) sharing economy businesses across the globe? How does the sharing economy adoption affect local regulations and stimulate free market policies in different nations?

we discuss in the next section.

4. A framework and research agenda for the sharing economy

There are important implications for theory and practice regarding the effects of the sharing economy on the global economy. The widespread adoption of platform-based businesses across the world is one of the fastest and largest internationalization movement to date. While industries are being overturned by these firms, the implications of this phenomenon need to be uncovered. There is still a lack of consensus about the competitive dynamics and internationalization patterns existing among C2C platforms, especially sharing economy firms. Drawing from previous IB literature, we noted that current theoretical knowledge on why, how, and when firms internationalize and compete overseas provides us with only a partial explanation of the behavior of internet-based firms such as sharing economy firms (Amit and Zott, 2001). On the other hand, this constitutes a promising avenue for future research.

We base our discussion on how future research could explore the sharing economy internationalization phenomenon by highlighting the importance of acknowledging business ecosystems across nations. Ecosystems are composed by all relevant stakeholders around a particular business and are constantly evolving as customers' preferences change and technological changes occur. In contrast to traditional global value chains (Gereffi et al., 2005; Porter, 1985), which are based on single-directed, location-bound flows of input from upstream providers to downstream consumers, ecosystems revolve around a platform or focal firm that connects suppliers and complementary asset providers from different industries and from different locations to consumer-participants in nonlinear ways (Adner, 2006; Adner and Kapoor, 2010; Iansiti and Levien, 2004). Thus, a key distinction of ecosystems versus global supply chains is its dependence on network effects and the nonlinearity of relationships in value-creating transactions (Thomas et al., 2014). Ecosystems are seen as open communities comprising different actors—such as direct suppliers, complementors, regulatory authorities, the judiciary system, and research institutions (Teece, 2007; Thomas et al., 2014)—that have different roles in the value creation and capture process.

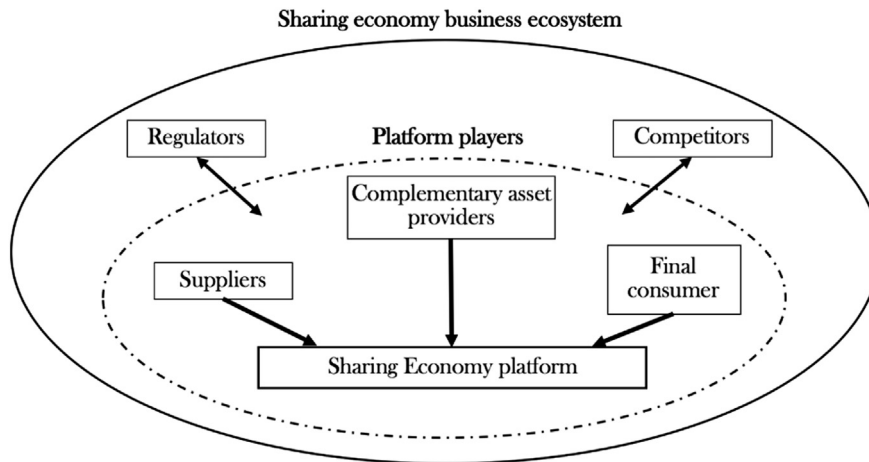


Fig. 3. Sharing economy platforms and its main business ecosystem's stakeholders.

Moreover, ecosystems also vary widely across nations due to different configurations of industry competitiveness, infrastructure, institutions, and consuetudes. An important benefit of utilizing the ecosystem perspective is its dynamic, not static, nature. According to Moore (1993), ecosystems evolve in life cycles as the interactions among stakeholders also change. In the early stages, firms and stakeholders are still adjusting and figuring out the best ways to create and capture value as new regulations are also implemented. In the later stages, the industry and regulations around it are established, and players have created barriers to entry due to technological leadership and widespread market adoption. In the business ecosystem perspective (see Fig. 3), industries are not isolated—business dynamics involve multiple supply chains that are widely impacted by cross-industry effects, which are usually related to technological advancements and innovation (Gulati et al., 2012; Moore, 1993). The ecosystem approach to sharing economy firms' internationalization should then encompass different IB perspectives such as locational advantages, and network and institutional effects as well as stakeholder dynamics, all embedded in the context of industry evolution.

As we have highlighted, since sharing economy firms do not possess assets to be transferred internationally and ownership advantages can be replicated by competitors, their internationalization relies mainly on their reach for multisided ecosystem players' engagement. According to former Uber executive Mina Radhakrishnan, the global market is seen as a “collection of hundreds of hyperlocal marketplaces,” where each local marketplace has its own ecosystem configuration. In this ecosystem view, the role of complementors from both downstream and upstream sides becomes extremely important in order to spur widespread adoption of the platform. Uber's partnership with important complementors—such as Toyota and the acquisition of maps start-up deCarta—reveal that sharing economy firms' network and integration investments prioritize complementary asset providers. This also indicates that these firms have to strategically “nurture” the ecosystem to enhance growth conditions for local providers, consumers, and complementors in different national ecosystems in order to survive (Rong et al., 2015). Such international ecosystem-based competitive strategies are still unassessed by the IB literature. The investigation of issues related to how sharing economy and other platforms foster multisided adoption and capture value in different geographies by organizing collaborative and knowledge-sharing partnerships is of great promise for the IB literature. While international network formation and configuration has an important role in firm internationalization (Choi and Contractor, 2016; Goerzen and Beamish, 2005; Zhou et al., 2007), little is known about how alliances and acquisitions made with players of different functions in local ecosystems (Kapoor and Furr, 2015; Rong et al., 2015; Wareham et al., 2014), and particularly with complementors, impact firm internationalization. Considering that business platforms need to be open to multisided players to capture value from transactions, open innovation (Baldwin and Hippel, 2011; Chesbrough, 2003) and collaborative innovation endeavors are also stimulated in ecosystems. Investigating sharing economy firms' internationalization using an ecosystem perspective would then allow for a more comprehensive view on key stakeholders and their important effects of firm international entry (Devinney et al., 2013).

Adner (2016) highlights that this “affiliation” perspective of ecosystems focuses on the breakdown of the supply and value chains, as network effects such as centrality and density are more emphasized. However, the author also points out that ecosystems should be seen as a “structure” in which players organize themselves in value creating activities. This structure perspective (Rong and Shi 2014) becomes relevant for the assessment of key value creating players and their impact on the viability of product/service being offered. Bringing the level of analysis to multisided relationships is particularly important in international business studies since platforms are more dependent on transactions with upstream, downstream and complementary asset partners. The focus of IB research on sharing economy firms should therefore be not only on how well embedded the firm is in the local ecosystem but also on how the firm can adjust its boundaries and create new value creating relationships as it expands into new markets. Next, we highlight opportunities and the importance of investigating sharing economy firms' internationalization and point out avenues of future research that are relevant for both theory and practice.

We start with a macro perspective particularly approaching the institutional and economic environment. By noticing that sharing economy firms have been successful in rapidly implementing their business model across the globe, it is important to note, however,

that in some locations, their activities have been banned permanently by local authorities. Although it is known that these firms usually enter a new market without fully complying with local regulations in order to gain local market support, there is a need to further understand how local institutions and sharing economy firms interact and eventually shape each other. Investigating the interactions between the local regulatory environment and sharing economy firms could result in important theoretical contributions from an institutional theory perspective by exploring how institutions are impacted and, at the same time, shape the path of disruptive technologies and market changes. Some pertinent unanswered questions are: (1) how policymakers should respond to sharing economy firms in order to further national competitiveness and welfare without compromising local firms; and (2) how the sharing economy adoption has affected local regulations and stimulated free market policies in different nations. Future research could also explore not only the regulatory changes necessary to govern the sharing economy sector, but also the cultural and societal aspects, such as trustworthiness and collectivism, shaped by increased adoption of sharing economy firms. The creation of a network-based marketplace by sharing economy firms also implies that multiple sectors of the economy are also impacted which, in turn, fosters innovativeness and allows individuals to compete against established firms not only locally but also in the international arena. Moreover, there is a need for research on the economical and social impacts brought by sharing economy firms to industries and nations, especially emerging economies. Questions related to what home country factors are conducive for the formation of sharing economy firms and how the dissemination of sharing economy firms is related to national competitiveness could provide relevant information to both theory and practice.

At the industry and firm levels, sharing economy firms have been able to engage users and providers into an efficient and transparent marketplace, increasing competition by offering an attractive alternative to consumers or even substituting established players. The investigation of how incumbent firms from various nations compete against sharing economy firms is important in order to shed light on how firms adjust their competitive strategies against new players relying on a collaborative platform-based business. Still, it is also important to investigate how sharing economy firms compete against incumbent players in different nations. Yet, investigations of the competitive dynamics of these firms are also missing in the literature. Research on sharing economy firms' international expansion strategies from an ecosystem-based perspective and the interplay between foreign and local sharing economy firms, as exemplified by the case of Uber and Didi in China, could provide important insights to the literature. Inquiries with regard to platform adaptation, trust building, management of complementary assets, and network developments could also advance research on MNEs' network-based internationalization strategy (Alcácer et al., 2016; Cantwell, 2013; Goerzen and Beamish, 2005). The investigation of sharing economy firms and their relationship management with complementary asset providers while entering foreign markets could also bring insights to current theoretical developments on entry mode (Eriksson et al., 1997; Hennart et al., 2015; Woodcock et al., 1994). Issues related to how sharing economy firms foster and nurture the local ecosystem utilizing different approaches for international locations could enlighten how these firms manage integration and responsiveness pressures utilizing a virtual platform and complementary asset relationships. Research on firms' internationalization using a business ecosystem perspective is still scarce (Rong et al., 2015), and future investigation could provide relevant insights to this important research stream.

Further understanding of how sharing economy firms develop competitive advantages in the local and global marketplaces is also necessary. As we highlighted, sharing economy firms may face issues related to the liability of foreignness and outsidership (Johanson and Vahlne, 2009) due to a lack of local legitimacy (Brouthers et al., 2016) and lack of understanding local customers' preferences. Departing from current knowledge on strategic and international management, future research should explore how sharing economy firms overcome these liabilities and develop and sustain competitive advantages without relying on asset ownership and internalized operations. Future investigations could provide important insights to internalization theory by addressing questions such as: (1) how sharing economy firms are able to sustain international competitiveness without relying on the internalization of foreign market activities; (2) how their multinationality-performance relationship is configured, given different institutional barriers and configurations (Geleilate et al., 2016; Kirca et al., 2011); (3) which dynamic capabilities these firms must develop and internalize to be successful internationally (Lessard et al., 2016); (4) how firms and managers deal with the opportunities and challenges associated with “gray” boundaries in the sharing economy; (5) What the implications for organization structures are (Birkinshaw and Morrison, 1995; Gupta and Govindarajan, 1991). The network-based business model of sharing economy firms also raises questions related to how these firms are able to control international activities given the low level of control it has on the user-provider interactions side (Brouthers et al., 2016; Eisenmann, 2006).

Given the increased importance of network effects for the viability of sharing economy firms, investigations focusing on how these firms acquire local legitimation and approval (Pant and Ramachandran, 2012), particularly when competing against local sharing economy firms, could enlighten network-based internationalization theories. The evidence provided that sharing economy firms focus on developing a virtuous network-based cycle in multiple international locations in order to be able to sustain their operations also raises questions on their real motivations to internationalize (Dunning, 1998). Yet, it is known that technology-based firms learn extremely fast and are able to efficiently process and learn from information sharing across the globe (Doz et al., 2001). Research on how sharing economy firms are able to convert these learnings and IT capabilities into competitive advantage against international and local players could also extend current knowledge on MNE capabilities development (Cantwell, 2014; Lessard et al., 2016). Specifically, little is known about how online data can best be captured to support MNEs' new product design and business model adaptation and how users contribute to and shape sharing economy platforms.

There are also potentially insightful investigations to be developed involving sharing economy firms at the individual level. Research on users' reactions and engagement toward the sharing economy in different national settings could provide important information to the understanding of how uncertainty avoidance, individualism, and other cultural characteristics interact with sharing-based businesses (Erez and Gati, 2004; Tomlinson, 2003). Investigating whether the adoption of sharing economy business has influenced users' perceptions of trust in other individual providers—both at home and when abroad—could be particularly

insightful. Moreover, the importance of entrepreneurs and their key role in sharing economy firms' competitive strategies and international expansion is also an important phenomenon that needs further research development.

In sum, we believe there is a great opportunity for future research on sharing economy firms' internationalization to extend the applications of most IB theories into new directions. This manuscript is only an incipient approach to a greater phenomenon, and more theoretical investigations are necessary to grasp more deeply the implications of the sharing economy phenomenon to global business. By raising awareness of the global economic, institutional, and societal impacts of the sharing economy, we hope to foster future research endeavors into investigating and enlightening the interesting behavior and repercussion of sharing economy firms.

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